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FGFR3 as a therapeutic target of the small molecule inhibitor PKC412 in hematopoietic malignancies

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Reccurent chromosomal translocation t(4;14) (p16.3;q32.3) occurs in patients with multiple myeloma (MM) and is associated with ectopic overexpression of fibroblast growth factor receptor 3 (FGFR3) that sometimes may contain the activation mutations such as K650E thanatophoric dysplasia type II (TDII). Although there have been significant advances in therapy for MM including the use of proteasome inhibitors, t(4;14) MM has a particularly poor prognosis and most patients still die from complications related to their disease or therapy. One potential therapeutic strategy is to inhibit FGFR3 in those myeloma patients that overexpress the receptor tyrosine kinase due to chromosomal translocation. Here we evaluated PKC412, a small molecule tyrosine kinase inhibitor, for treatment of FGFR3-induced hematopoietic malignancies. PKC412 inhibited kinase activation and proliferation of hematopoietic Ba/F3 cells transformed by FGFR3 TDII or a TEL-FGFR3 fusion. Similar results were obtained in PKC412 inhibition of several different t(4;14)-positive human MM cell lines. Furthermore, treatment with PKC412 resulted in a statistically significant prolongation of survival in murine bone marrow transplant models of FGFR3 TDII-induced pre-B cell lymphoma, or a peripheral T-cell lymphoma associated TEL-FGFR3 fusion-induced myeloproliferative disease. These data indicate that PKC412 may be a useful molecularly targeted therapy for MM associated with overexpression of FGFR3, and perhaps other diseases associated with dysregulation of FGFR3 or related mutants.
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Introduction

Multiple myeloma (MM) is characterized by clonal expansion of terminally differentiated plasma B cells and is among the most common hematologic malignancies in patients over the age of 65 years. Recurrent chromosomal translocations of 14q32 into the immunoglobulin heavy (IgH) chain switch region are frequently identified in human MM cells (Bergsagel *et al*., 1996), and result in dysregulated expression of several heterogeneous partners including cyclin D1 (Chesi *et al*., 1996), c-maf (Chesi *et al*., 1998a) and FGFR3 (Chesi *et al*., 1997). The t(4;14) involving FGFR3 has been identified in approximately 15% of MM patients and in cell lines derived from patients with MM (Chesi *et al*., 1997, 1998b). In some cases, the translocated *FGFR3* gene contains a K650E mutation that constitutively activates the FGFR3 kinase, and when present in the germ line, causes thanatophoric dysplasia type II (TDII) (Tavormina *et al*., 1995).

FGFR3 belongs to the type III receptor-tyrosine kinase family that responds to fibroblast growth factor (FGF), and negatively regulates bone formation in mammals (Colvin *et al*., 1996). FGF binds to the extracellular domain of FGFR3 and results in receptor oligomerization, which leads to autophosphorylation at tyrosine residues in the cytoplasmic domains. The phosphorylated tyrosine residues are required for either stimulation of the intrinsic catalytic activity or activation of downstream signaling pathways by providing docking sites for SH2 domain containing signaling components (Webster and Donoghue, 1996). FGFR3 is oncogenic when activated by FGF ligands or by activating mutations. For example, overexpression of FGFR3 wild-type or activated FGFR3 TDII mutant transforms murine B9 myeloma cells to IL-6-independent

growth (Plowright *et al*., 2000). The activated form of FGFR3 induces transformation of NIH3T3 cells that are tumorigenic when injected into nude mice (Chesi *et al*., 2001). Moreover, in a murine bone marrow transplantation (BMT) model, mice transplanted with bone marrow cells transduced by retroviral vectors, encoding the activated FGFR3 TDII mutant, rapidly develop a lethal pre-B cell lymphoma (Li *et al*., 2001).

In addition, dysregulation of FGFR3 has been associated with other hematopoietic malignancies. For example, a chromosomal translocation t(4;12) (p16;p13) was reported in one case of human peripheral T-cell lymphoma. The translocation results in expression of a fusion TEL-FGFR3 tyrosine kinase with the N-terminal pointed (PNT) domain of a transcription factor TEL (ETV6) fused to the C-terminal intracellular tyrosine kinase domain of FGFR3 (Yagasaki *et al*., 2001). The TEL PNT domain mediates self-association of the fusion protein, and results in constitutive activation of TEL-FGFR3, as has been reported for other TEL-tyrosine kinase fusions such as TEL-PDGFβR (Golub *et al*., 1996).

The t(4;14) associated MM has a particularly poor diagnosis. Although there have been significant recent advances in treatment of MM, most patients still die from complications related to their disease or therapy. The median survival of patients treated with conventional chemotherapy is approximately 3–4 years (San Miguel *et al*., 1999). Some therapies for MM, including high-dose chemo- or radio-therapy followed by autologous transplantation of hematopoietic stem cells, may improve clinical outcome and enhance survival. However, most patients are not suitable for such aggressive approaches and only 20–40% of patients receiving these therapies achieve stringently-defined complete remission and most of them subsequently relapse (Cavo *et al*., 2000; Lemoli *et al*., 2000; Martinelli *et al*., 2000). Among the newer treatments, the antiangiogenic agent thalidomide (Richardson *et al*., 2002; Heffner and Lonial, 2003) and the proteasome inhibitor Bortezomib (Paramore and Frantz, 2003) are valuable additions to the pharmacologic armamentarium to treat MM, but are not curative therapies as single agents and may be associated with significant side effects in some patients.

We hypothesized that inhibition of FGFR3 could be an effective therapeutic strategy in treatment of human MM and other hematopoietic malignancies involving dysregulation of FGFR3. In this report, we tested the potential therapeutic utility of the small molecule kinase inhibitor PKC412 as an option for treatment of leukemias associated with activated FGFR3 mutants. PKC412 is a potent inhibitor of several kinases including FLT3, PKC, KDR, c-KIT, PDGFRα and PDGFβR (Andrejauskas-Buchdunger and Regenass, 1992; Fabbro *et al*., 2000; Weisberg *et al*., 2002), and is currently being evaluated in Phase II clinical trials for AML patients with and without FLT3 activating mutations (Estey *et al*., 2003; Stone *et al*., 2003). Small molecule inhibitors of FGFR3 such as PD173074 and SU5402 have been shown to effectively inhibit FGFR3 in t(4;14)-positive primary MM cell lines including KMS11, KMS18 and OPM-2, as well as in a xenograft murine model (Grand *et al*., 2004; Paterson *et al*., 2004; Trudel *et al*., 2004). However, unlike PKC412, none of them has been reported in the literature, to our knowledge, to be efficacious and well tolerated in patients. We have demonstrated that PKC412 also has inhibitory activity for FGFR family members such as FGFR1 (Chen *et al*., 2004a), and was therefore tested for activity to inhibit FGFR3 TDII mutant, as well as a TEL-FGFR3 fusion tyrosine kinase *in vitro* and *in vivo*.

Results

PKC412 inhibits FGFR3 TDII and the TEL-FGFR3 fusion tyrosine kinase in hematopoietic Ba/F3 cells

We evaluated the transforming properties of FGFR3 TDII (K650E) mutant and TEL-FGFR3 fusion tyrosine kinase (Figure 1a) in murine hematopoietic Ba/F3 cells. Both FGFR3 mutants were subcloned into retroviral vectors carrying a neomycin resistance gene and stably transduced in Ba/F3 cells. Stable Ba/F3 cell lines were assessed for IL-3-independent growth as a surrogate for transformation. FGFR3 TDII mutant is constitutively activated and can be further activated in the presence of ligand (Naski *et al*., 1996). Indeed, in the presence of acidic FGF (aFGF) ligand, FGFR3 TDII effectively conferred IL-3-independence to Ba/F3 cells, whereas control Ba/F3 cells transduced with empty vector underwent apoptosis in the absence of IL-3. In contrast, TEL-FGFR3 effectively transformed Ba/F3 cells to factor-independent growth, independent of aFGF ligand (Figure 1b). The increased ability of the TEL-FGFR3 fusion to confer factor-independent growth of Ba/F3 cells may be due to a higher expression level of TEL-FGFR3 in the stably transduced Ba/F3 cells when compared with FGFR3 TDII protein levels (Figure 1b).

Previous enzymatic assays had demonstrated inhibitory activity of PKC412 for FGFR family members, and we have demonstrated that PKC412 is a potent inhibitor to ZNF198-FGFR1 *in vitro* and *in vivo* (Chen *et al*., 2004a). To determine whether PKC412 inhibited FGFR3-activated mutants, we performed a dose–response analysis of Ba/F3 cells stably expressing FGFR3 TDII or the TEL-FGFR3 fusion. PKC412 effectively inhibited the growth of FGFR3 TDII-and TEL-FGFR3-transformed Ba/F3 cells in the absence of IL-3 with a cellular IC_{50} of $\sim$240 and 200 nM, respectively (Figure 1c). Since PKC412 inhibits several kinases, we next tested whether mutant FGFR3 was the critical target for PKC412-mediated cytotoxicity in this assay. We have demonstrated previously that PKC412 does not have nonspecific cytotoxicity in Ba/F3 cells growing in the presence of IL-3 at concentrations less than 500 nM (Cools *et al*., 2003; Chen *et al*., 2004a). However, to determine whether these FGFR3 mutants were the critical targets for PKC412-mediated cytotoxic effects, we introduced an N540D mutation in the context of native FGFR3 into FGFR3 TDII and the TEL-FGFR3 fusion. N540D corresponds to a N659D

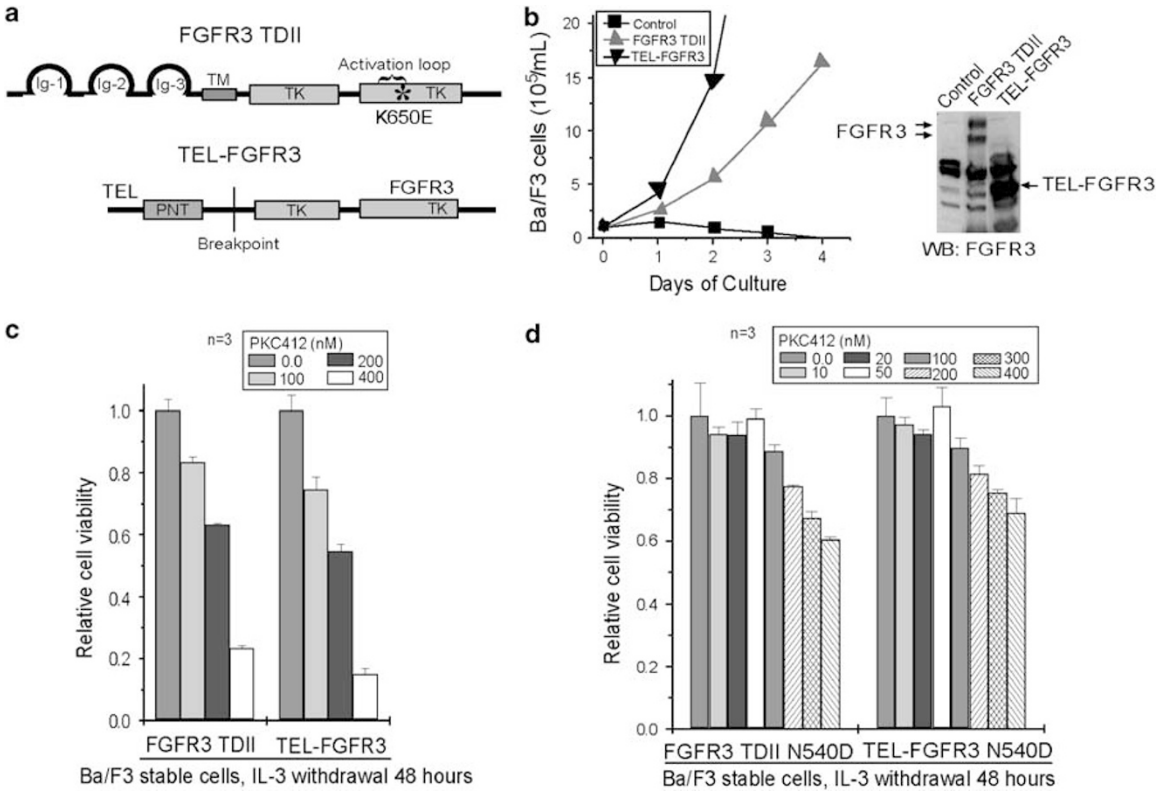

Figure 1 PKC412 inhibits FGFR3 TDII and TEL-FGFR3 in Ba/F3 cells. (**a**) Schematic diagram of FGFR3 and TEL-FGFR3 fusion tyrosine kinase. (**b**) FGFR3 TDII and TEL-FGFR3 confer IL-3-independence to Ba/F3 cells. FGFR3 TDII and TEL-FGFR3 were stably transduced into Ba/F3 cells. Left panel: cells were cultured in absence of IL-3 and counted daily; cells expressing FGFR3 TDII mutant were treated with aFGF ligand during the experiment. Ba/F3 cells transduced by an empty retroviral vector were included as a control. Result presented is a representative of three independent experiments. Right panel: protein expression level of FGFR3 TDII and TEL-FGFR3 in stably transduced Ba/F3 cells. (**c**) Dose–response analysis of Ba/F3 cells stably expressing FGFR3 TDII or TEL-FGFR3 fusion protein. Cells were treated with PKC412 and IL-3 withdrawal for 48 h prior to analysis in a cell viability assay. Cells expressing FGFR3 TDII mutant were treated with aFGF ligand during the experiment. The relative cell viability was normalized to the viability of cells in the absence of the drug. (**d**) N540D mutation in the context of FGFR3 confers resistance to PKC412. Ba/F3 cells transduced with FGFR3 TDII N540D mutant or TEL-FGFR3 N540D mutant were tested for growth with treatment of PKC412 in the absence of IL-3

mutation in the context of PDGFRα that confers PKC412-resistance to the FIP1L1-PDGFRα fusion tyrosine kinase (Cools *et al*., 2003). We hypothesized that N540D in the context of the related RTK type III family member FGFR3 would also confer PKC412 resistance. Indeed, Ba/F3 cells transformed with FGFR3 TDII K650E/N540D or TEL-FGFR3 N540D demonstrated resistance to PKC412, with an elevated cellular IC_{50} that was not reached at PKC412 concentration as high as 400 nM (Figure 1d). These data confirm that N540D confers resistance to PKC412, and that these FGFR3 mutants are the critical targets for PKC412-mediated cytotoxicity. These findings are corroborated by data showing that PKC412 effectively inhibited TEL-FGFR3 tyrosine autophosphorylation, as well as phosphorylation of known FGFR3 signaling intermediates including STAT5, PLCγ and PI3K (Figure 2).

Inhibition of t(4;14)-positive primary MM cell lines by PKC412

Next we tested the inhibitory effects of PKC412 in a more clinically relevant system by performing a dose–response analysis using t(4;14)-positive primary MM cell lines, including OPM-1, LP1 and KMS11. OPM-1 and KMS11 cells overexpress FGFR3 mutant with a single activating mutation K650E in the kinase domain and Y373C in the transmembrane domain (Ronchetti *et al*., 2001), respectively, whereas LP1 cells express FGFR3 wild-type with a polymorphism of F384L in the transmembrane domain (Golla *et al*., 1997). PKC412 effectively inhibited the growth of OPM-1 and KMS11 cells (Figure 3a). PKC412 also inhibits LP1 cells with relatively higher cellular IC_{50}, and such difference of drug-sensitivity might be due to the much less expression level of FGFR3 protein in LP1 cells than in OPM-1 cells (Figure 3a and b). In contrast, no significant inhibition was observed in a t(4;14)-negative MM cell line OCI-My5 in which no expression of FGFR3 was detected (cellular $IC_{50} > 5\ \mu M$) (Figure 3a and b).

These data suggest that PKC412 effectively inhibits proliferation of FGFR3-positive t(4;14) MM cells and that FGFR3 is the molecular target for inhibition of the t(4;14) MM cells by PKC412. Moreover, PKC412 has minimal nonspecific cytotoxicity in all tested MM cell lines and does not inhibit the control t(4;14)-negative

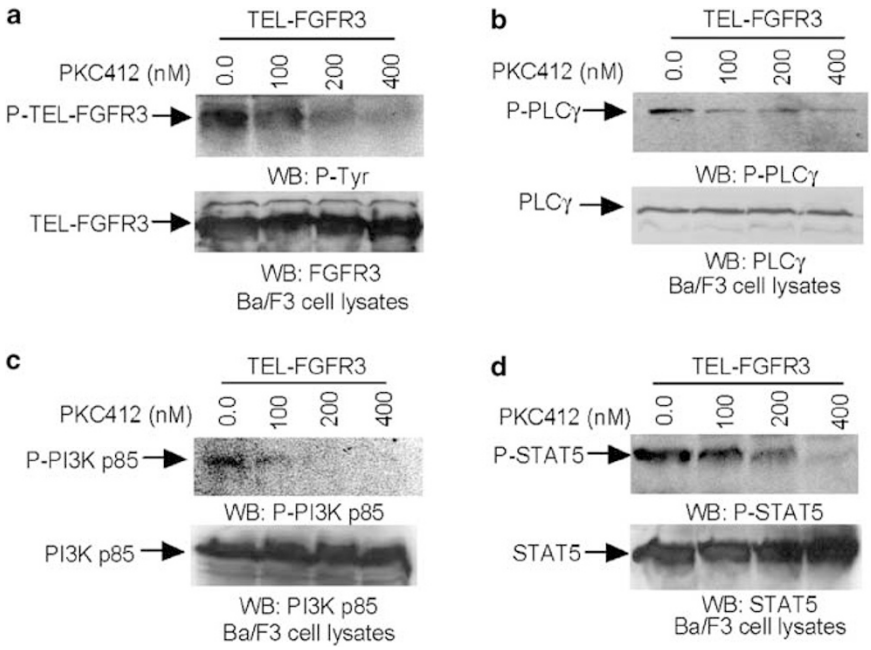

Figure 2 (**a**) PKC412 inhibits autophosphorylation of TEL-FGFR3. Ba/F3 cells stably expressing TEL-FGFR3 were cultured in the presence of increasing concentrations of PKC412. Tyrosine-phosphorylated FGFR3 was assessed by using antiphosphotyrosine antibody 4G10, whereas control TEL-FGFR3 protein was probed by a FGFR3 antibody. (**b**–**d**) Analysis of phosphorylation status of TEL-FGFR3 downstream signaling components STAT5, PLCγ and PI3K. Immunoblotting was performed using whole cell lysates of Ba/F3 cells that were incubated with increasing concentrations of PKC412

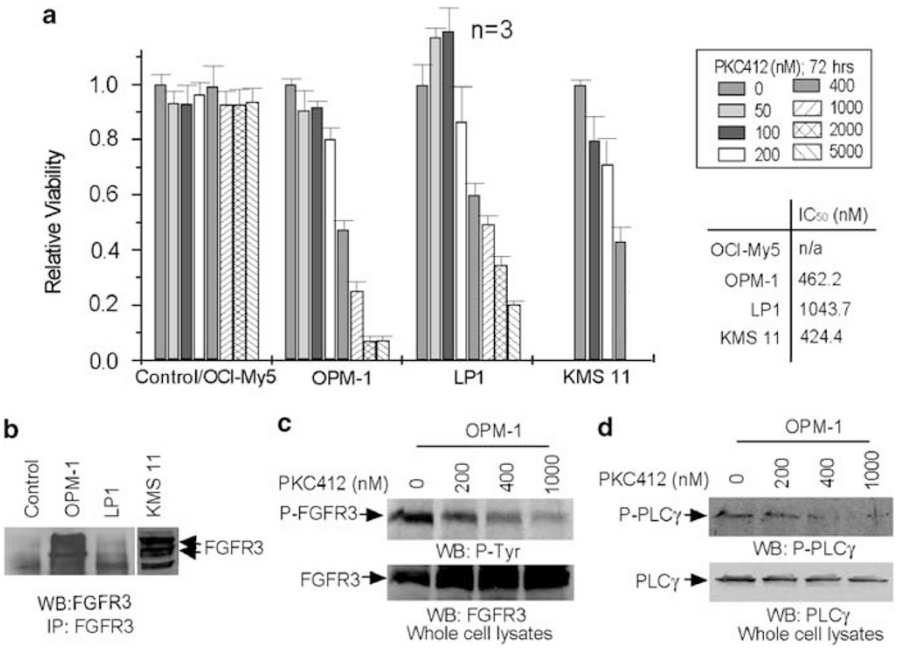

Figure 3 PKC412 inhibits t(4;14) MM cell lines. (**a**) Dose–response analysis of t(4;14)-positive primary MM cell lines to PKC412 treatment. Left panel: cells were treated with PKC412 for 72 h prior to analysis in a cell viability assay. The relative cell viability was normalized to the viability of cells in the absence of the drug. A t(4;14)-negative primary myeloma cell line, OCI-My5 was included as a negative control. Right panel shows the cellular IC_{50} (nM) of distinct cell lines tested. (**b**) Protein expression of FGFR3 in t(4;14)-positive MM cell lines OPM-1, LP1 and KMS11. FGFR3 immunocomplexes were obtained and blotted with anti-FGFR3 antibody. OCI-My5 cells were included as a negative control. (**c**) PKC412 inhibits autophosphorylation of FGFR3 in t(4;14)-positive OPM-1 cells. Cells were cultured in the presence of increasing concentrations of PKC412 and tyrosine-phosphorylated FGFR3 was assessed by 4G10. Lower panel shows the total FGFR3 protein in each sample probed by an FGFR3 antibody. (**d**) Tyrosine phosphorylation of FGFR3 downstream signaling component PLCγ. Immunoblotting was performed using whole-cell lysates of OPM-1 cells that were incubated with increasing concentrations of PKC412

OCI-My5 MM cells. These data also correlate with the observations that PKC412 effectively inhibited FGFR3 tyrosine autophosphorylation as well as phosphorylation and activation of signaling intermediate such as PLCγ in OPM-1 cells (Figure 3c and d).

PKC412 is effective for the treatment of FGFR3 TDII-induced immature B-cell lymphoma in a murine BMT model

The murine BMT disease model of FGFR3 TDII-induced pre-B cell lymphoma (Li *et al*., 2001) suggested a model system to test the efficacy of PKC412 for treatment of disease induced by FGFR3 TDII. We performed the murine BMT assay using a murine bicistronic retroviral vector containing FGFR3 TDII cDNA and an IRES-EGFP expression cassette. Animals receiving bone marrow cells transduced by FGFR3 TDII rapidly developed a fatal immature B-cell lymphoma as reported. The FGFR3 TDII transplanted mice were divided into two groups that were treated with daily oral gavage of placebo or PKC412, respectively. Animals in the placebo-treated control group developed pre-B cell lymphoma as observed previously, characterized by splenomegaly, lymphadenopathy and marked leukocytosis, and were all killed by day 25 post-transplantation (Figure 4a and Table 1). The PKC412 group was treated for one additional week after all the placebo-treated animals had been killed for progressive disease. There was a statistically significant prolongation of survival in the PKC412 treated group, with seven out of eight mice alive at the end point of the study (day 32, $P = 0.0006$). PKC412-treated animals also had markedly reduced spleen weights and white blood cell (WBC) counts that were statistically significantly different from placebo-treated animals (Figure 4a and Table 1).

Efficacy of PKC412 treatment was confirmed by histopathologic examination that demonstrated a marked reduction of lymphoid disease in multiple organs including bone marrow, spleen and liver in drug-treated animals. These findings were corroborated by flow cytometric analysis of bone marrow cells from a representative PKC412-treated mouse which displayed a marked decrease of the immature B-cell population (1%; B220+, CD19+) when compared with marrow from a placebo-treated mouse (73%) (Figure 4b and c). These data indicate that PKC412 is efficacious in the treatment of immature B-cell lymphoid disease induced by FGFR3 TDII in a murine BMT model.

Efficacy of PKC412 in treatment of TEL-FGFR3-induced myeloproliferative disease in the murine BMT model

We also tested the *in vivo* transforming activity of the TEL-FGFR3 fusion in primary hematopoietic cells in a murine BMT assay. Unlike the FGFR3 TDII BMT model, animals receiving bone marrow cells transduced by TEL-FGFR3 developed a rapidly fatal myeloproliferative disease characterized by marked splenomegaly and a peripheral blood leukocytosis comprised predominantly of mature granulocytes (Figure 5a–c and

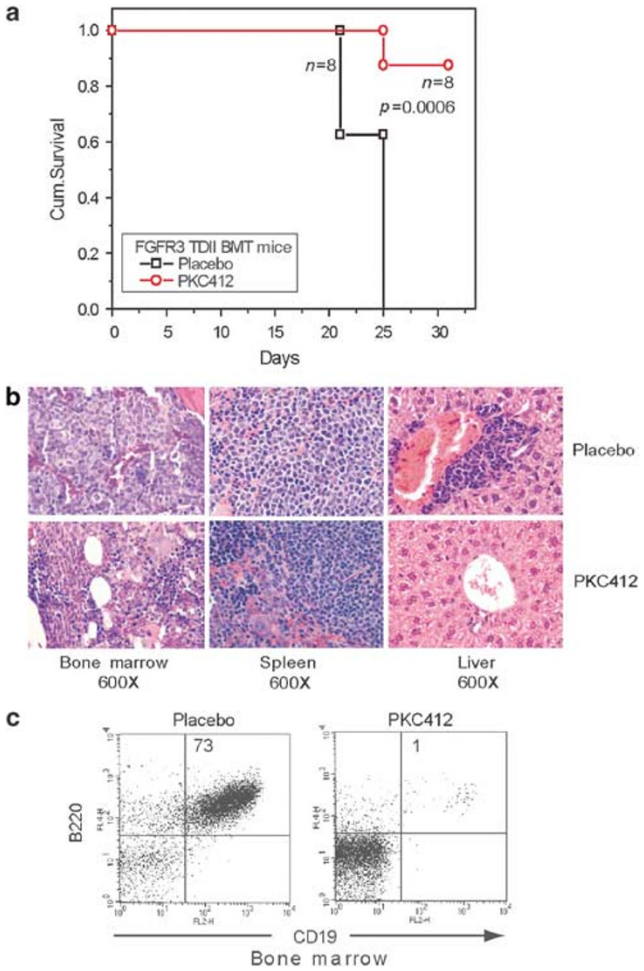

Figure 4 PKC412 increases survival in a murine BMT model of a pre-B-cell lymphoma induced by FGFR3 TDII. (**a**) Kaplan–Meier plot of the survival of mice transplanted with bone marrow cells expressing FGFR3 TDII that were treated with either placebo or PKC412. (**b**) Histopathology of bone marrow, the spleen and liver of placebo- and PKC412- treated FGFR3 TDII mice at the time of killing or study end point. A markedly reduced population of immature lymphoid elements was observed in the multiple tissues of PKC412-treated mice compared with placebo-treated animals. (**c**) Flow cytometry analysis showed a marked reduction of B220/CD19 double-positive cells in the bone marrow of PKC412-treated FGFR3 TDII mice

Table 1 Efficacy of PKC412 in the treatment of FGFR3 TDII-induced pre-B-cell lymphoma or TEL-FGFR3-induced myeloproliferative disease

	Placebo	*PKC412*
FGFR3 TDII (n = 8)		
Spleen weight (mg)/median	430.7–673.6/591.65	23.4–257.1/144.75
WBC count (10^6/ml)/median	19–101/77.2	1.4–3.77/2.42
TEL-FGFR3 (n = 8)		
Spleen weight (mg)/median	666–1241/905.5	97–223/126.5
WBC count (10^6/ml)/median	44–202/82.5	1–33/4.5

n = number of animals in each group

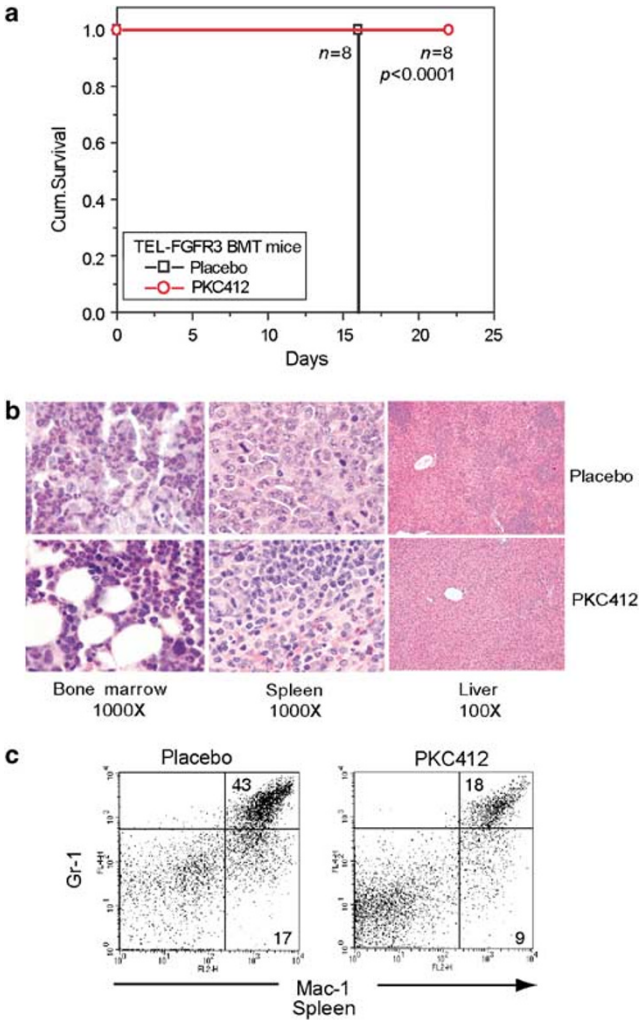

Figure 5 PKC412 increases survival of mice with a myeloproliferative disease induced by TEL-FGFR3 in a murine BMT assay. (**a**) Kaplan–Meier plot of the survival of TEL-FGFR3-transplanted mice that were treated with either placebo or PKC412. (**b**) Histopathologic examination of tissue sections of placebo- and PKC412- treated TEL-FGFR3 mice. Marked predominance of maturing myeloid elements was observed in the bone marrow as well as extensive amounts of extramedullary hematopoiesis in both the liver and spleen comprised of maturing myeloid cells from placebo-treated mice. Sections from PKC412-treated animals displayed maturing trilineage hematopoiesis in the bone marrow and a significant reduction of extramedullary hematopoiesis in the spleen and liver. (**c**) Flow cytometry analysis showed a marked reduction of Gr-1/Mac-1-positive cells in the spleen of a PKC412-treated representative TEL-FGFR3 mouse, compared with a control placebo-treated TEL-FGFR3 mouse

Table 1). Such a difference in phenotype may be due to differences in intracellular localization of the two proteins as full-length FGFR3 TDII is a membrane protein whereas TEL-FGFR3 fusion is probably localized in the cytosol. Histopathologic examination of TEL-FGFR3-transplanted mice demonstrated perturbation of normal splenic architecture with loss of white pulp and expansion of the red pulp by a prominent population of maturing myeloid forms (Figure 5b). Bone marrow samples from TEL-FGFR3 mice were markedly hypercellular with a predominance of mature myeloid forms. Also noted were a frequent number of admixed histiocytes and macrophages. Flow cytometric analysis showed abnormally elevated numbers of mature neutrophils that were positive for the late myeloid markers Gr-1 and Mac-1 in spleen samples of TEL-FGFR3-transplanted animals (Figure 5c).

We next tested the efficacy of PKC412 in the treatment of TEL-FGFR3-induced myeloproliferative disease in mice. We repeated the BMT assay and the mice were treated with daily doses of placebo or PKC412 by oral gavage. As observed previously, placebo-treated mice developed myeloproliferative disease, whereas the PKC412-treated group had a statistically significant prolongation in survival ($P<0.0001$), as well as a marked reduction in spleen size and peripheral WBC counts (Figure 5a and Table 1). Histopathologic examination showed a significant reduction of maturing myeloid elements in the spleens of PKC412-treated animals when compared with placebo-treated animals (Figure 5b), which was corroborated by flow cytometric analysis that demonstrated a marked decrease in the percentage of Gr-1/Mac-1 double-positive cells in the spleen from a representative PKC412-treated mouse (18%) compared with a representative placebo-treated mouse (43%) (Figure 5c).

Discussion

Small molecule tyrosine kinase inhibitors, such as imatinib (Gleevec) or gefitinib (Iressa), are efficacious in treating certain human malignancies and solid tumors associated with dysregulation of tyrosine kinases. For example, imatinib has been successfully applied in clinical treatments of BCR-ABL-associated chronic myelogenous leukemia (CML), TEL-PDGFβR-induced chronic myelomonocytic leukemia (CMML), FIP1L1-PDGFRA-associated hypereosinophilic syndrome and activating mutations of c-KIT-associated gastrointestinal stromal cell tumors (GIST) (Apperley *et al*., 2002; Demetri *et al*., 2002; Cools *et al*., 2003; Druker, 2003). PKC412 is being evaluated in clinical phase I/II trials for treatment of AML associated with wildtype or activating mutant forms of FLT3 (Weisberg *et al*., 2002; Estey *et al*., 2003). Recently, PKC412 has been shown to inhibit FGFR1. For example, we have recently demonstrated that PKC412 is an effective inhibitor of the ZNF198-FGFR1 fusion associated with stem-cell myeloproliferative disorder *in vivo* and *in vitro*, and have observed clinical efficacy of PKC412 in treatment of a patient with ZNF198-FGFR1-positive myeloproliferative disease (MPD) (Chen *et al*., 2004a).

In this report, we evaluated PKC412 as an inhibitor of activating FGFR3 mutants. We observed that PKC412 effectively inhibits FGFR3 TDII- and TEL-FGFR3-dependent transformation of Ba/F3 cells, as well as kinase activity and activation of downstream signaling effectors. We generated PKC412-resistant (N540D) FGFR3 mutants, thereby demonstrating that although

PKC412 inhibits several different kinases, the critical target for PKC412-induced cytotoxicity was the activated FGFR3 in hematopoietic Ba/F3 cells in these cell assays (Figure 1d). Moreover, similar results were obtained in inhibition of t(4;14)-positive human MM cell lines by PKC412. Interestingly, the sensitivity to PKC412 is different among FGFR3-positive MM cell lines including OPM-1, LP1 and KMS11, which might be due to variant FGFR3 expression levels in the cells. However, we cannot exclude the possibility that there might be other oncogenic abnormalities that are not responsive to PKC412 treatment in LP1 MM cells in addition to FGFR3. Larger numbers of cell lines will need to be evaluated to determine efficacy of PKC412 in other MM cell lines that overexpress FGFR3.

In consonance with the data obtained in biochemical and cell culture assays, PKC412 was efficacious in treating both FGFR3-TDII-induced lymphoid disease and TEL-FGFR3-induced myeloproliferative disease in the murine BMT models. Taken together, our data suggest that pharmacologic inhibition of FGFR3 activating mutants associated with hematologic malignancy with small molecule inhibitors like PKC412 may be efficacious in clinical treatment of t(4;14)-associated MM, and perhaps for other human hematopoietic disorders associated with dysregulated FGFR3 activity, including t(4;12) (p16;p13)-associated TEL-FGFR3-induced peripheral T-cell lymphoma.

Previous studies have also demonstrated that the small molecule inhibitors of FGFR3 such as PD173074 and SU5402 effectively inhibits t(4;14)-positive primary MM cell lines including KMS11, KMS18 and OPM-2 in cell culture, as well as in a xenograft murine model (Grand *et al*., 2004; Paterson *et al*., 2004; Trudel *et al*., 2004). The strength in the utility of PKC412 as a novel small molecule inhibitor in the treatment of EGFR3-associated multiple myeloma is that PKC412 is currently being evaluated in human clinical trials for treatment of AML associated with FLT3, whereas no other FGFR3 inhibitors have been reported in the literature, to our knowledge, to be efficacious and well tolerated in patients. Thus far, PKC412 has shown minimal side effects in phase I/II clinical trials. This brings much promise to MM patients who receive current highly aggressive therapies such as high-dose chemo- or radiotherapy followed by autologous transplantation of hematopoietic stem cells, and who suffer from significant side effects such as peripheral neuropathy caused by thalidomide (Bastuji-Garin *et al*., 2002).

On the other hand, in addition to activation of *FGFR3*, the t(4;14) (p16;q32) also results in creation of a chimeric fusion transcript between *IGH* and *MMSET* (MM SET domain) (Chesi *et al*., 1998b). A recent report demonstrates that overexpression of transcripts originating from *MMSET* is associated with all t(4;14) (p16;q32)-positive MM patients (Keats *et al*., 2005), and approximately 32% of the newly diagnosed cases of t(4;14) (p16;q32) MM express the *IGH-MMSET* fusion transcript but lack expression of FGFR3 (Santra *et al*., 2003). This distinction suggests that FGFR3 may not be the only relevant gene for pathogenesis in t(4;14) (p16;q32) MM, and that *IGH-MMSET* expression could conceivably modulate the potential therapeutic benefit of PKC412 in treatment of t(4;14) (p16;q32) MM.

Activating mutations of FGFR3 such as kinase domain mutation K650E, as well as extracellular domain mutations R248C, S249C and G370C, have also been identified in human bladder and cervical carcinomas (Cappellen *et al*., 1999). Therefore, our findings that the small molecule tyrosine inhibitor PKC412 inhibits FGFR3 activating mutants may have therapeutic implications in the treatments of not only various hematopoietic malignancies but also solid tumors associated with dysregulated FGFR3.

Materials and methods

DNA constructs

The full-length FGFR3 TDII (K650E) cDNA and TEL-FGFR3 fusion cDNA were subcloned into the retroviral destination vectors MSCV-Gateway-neoEB and MSCV2.2-Gateway-IRESGFP as described (Chen *et al*., 2005). All the constructs generated were confirmed by DNA sequencing.

Cell culture and retrovirus production

Ba/F3 cells were cultured in RPMI 1640 medium in presence of 10% fetal bovine serum (FBS) and 1.0 ng/ml, interleukin-3 (IL-3) (R&D Systems, Minneapolis, MN, USA). Primary MM cell lines were culture in RPMI 1640 medium with 10% FBS. A total of 293 T cells were cultured in Dulbecco modified Eagle medium (DMEM) with 10% FBS. The retroviral stocks were generated and the viral titers were determined as described previously (Schwaller *et al*., 1998; Liu *et al*., 2000). For the murine BMT experiments, the viral titers of all constructs were normalized to $\sim 1 \times 10^6$ infectious units/ml. Stable neomycin-resistant Ba/F3 cell lines expressing FGFR3 TDII or TEL-FGFR3 were generated and the IL-3-independent proliferation assay was performed as described (Chen *et al*., 2004b). For cell viability assay, 1×10^5 Ba/F3 cells or primary MM cells were cultured in 24-well plates under condition of IL-3 withdrawal and increasing concentrations of PKC412. The number of viable cells at each experimental time point was determined by using the Celltiter96AQ$_{\text{uesous}}$one solution proliferation kit (Promega, Madison, Wisconsin, USA). The FGFR3 TDII stable cells were treated with acidic FGF (1.0 nM) and heparin (30 μg/ml) during the cell proliferation assays.

Western blot

When assayed for phosphorylation level of different protein factors, Ba/F3 cells and primary MM cell lines were treated with serum starvation and, in some instances, PKC412 for 4 h prior to lysis. The cell extracts were clarified by centrifugation. The enzyme-linked immunoblotting procedures were performed essentially as described (Sternberg *et al*., 2001). Crossreacting materials were visualized by enhanced chemiluminescence (ECL). Applied antibodies include: polyclonal rabbit anti-FGFR3, rabbit polyclonal antibodies against STAT5b, phospho-PI3K-p85 (Tyr-508) (Santa Cruz Biotechnology, Santa Cruz, CA, USA); mouse monoclonal 4G10 against phosphotyrosine (Upstate Biotechnology, Lake Placid, NY, USA); rabbit polyclonal antibodies against phospho-

STAT5 (Tyr-694), PLC-γ1, phospho-PLC-γ1 (Tyr-783) (Cell Signaling, Beverly, MA, USA).

Murine BMT assay and PKC412 treatment of the animals

The murine BMT assays and drug treatment were performed as described previously (Persons *et al*., 1997; Liu *et al*., 2000; Kelly *et al*., 2002; Cools *et al*., 2003). In brief, 1×10^6 bone marrow cells transduced with distinct retroviral constructs were injected into the lateral tail veins of lethally irradiated (450 cGy $\times$ 2) syngeneic Balb/c recipient mice. PKC412 was prepared as 6% w/w stock in Gelucire® 44/14 (GC) (Gattefosse, France) and diluted with sterile deionized water prior to administration. Dosing of PKC412 (100 mg/kg/day) was performed every 24 h by oral gavage, as previously described (Cools *et al*., 2003). Placebo group of mice received equal volume of GC solution. Diseased animals were identified by splenomegaly or moribund appearance and were killed for further analyses. Histopathological analyses were performed; single-cell suspensions of bone marrow, spleen and peripheral blood were analysed by flow cytometric analysis as described previously (Schwaller *et al*., 1998). Statistical significance for survival analysis was assessed using the log-rank test.

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References

Andrejauskas-Buchdunger E and Regenass U. (1992). *Cancer Res.*, **52,** 5353–5358.
Apperley JF, Gardembas M, Melo JV, Russell-Jones R, Bain BJ, Baxter EJ, Chase A, Chessells JM, Colombat M, Dearden CE, Dimitrijevic S, Mahon FX, Marin D, Nikolova Z, Olavarria E, Silberman S, Schultheis B, Cross NC and Goldman JM. (2002). *N. Engl. J. Med.*, **347,** 481–487.
Bastuji-Garin S, Ochonisky S, Bouche P, Gherardi RK, Duguet C, Djerradine Z, Poli F and Revuz J. (2002). *J. Invest. Dermatol.*, **119,** 1020–1026.
Bergsagel PL, Chesi M, Nardini E, Brents LA, Kirby SL and Kuehl WM. (1996). *Proc. Natl. Acad. Sci. USA*, **93,** 13931–13936.
Cappellen D, De Oliveira C, Ricol D, de Medina S, Bourdin J, Sastre-Garau X, Chopin D, Thiery JP and Radvanyi F. (1999). *Nat. Genet.*, **23,** 18–20.
Cavo M, Terragna C, Martinelli G, Ronconi S, Zamagni E, Tosi P, Lemoli RM, Benni M, Pagliani G, Bandini G and Tura S. (2000). *Blood*, **96,** 355–357.
Chen J, DeAngelo DJ, Kutok JL, Williams IR, Lee BH, Wadleigh M, Duclos N, Cohen S, Adelsperger J, Okabe R, Coburn A, Galinsky I, Huntly B, Cohen P, Meyer T, Fabbro D, Banerji L, Griffin JD, Xiao S, Fletcher JA, Stone RM and Gilliland DG. (2004a). *Proc. Natl. Acad. Sci. USA*, **101,** 14479–14484.
Chen J, Williams IR, Kutok JL, Duclos N, Anastasiadou E, Masters SC, Fu H and Gilliland DG. (2004b). *Blood*, **104,** 535–542.
Chen J, Williams IR, Lee BH, Duclos N, Huntly B, Donoghue DJ and Gilliland DG. (2005). *Blood*, **106,** 328–337.
Chesi M, Bergsagel PL, Brents LA, Smith CM, Gerhard DS and Kuehl WM. (1996). *Blood*, **88,** 674–681.
Chesi M, Bergsagel PL, Shonukan OO, Martelli ML, Brents LA, Chen T, Schrock E, Ried T and Kuehl WM. (1998a). *Blood*, **91,** 4457–4463.
Chesi M, Brents LA, Ely SA, Bais C, Robbiani DF, Mesri EA, Kuehl WM and Bergsagel PL. (2001). *Blood*, **97,** 729–736.
Chesi M, Nardini E, Brents LA, Schrock E, Ried T, Kuehl WM and Bergsagel PL. (1997). *Nat. Genet.*, **16,** 260–264.
Chesi M, Nardini E, Lim RS, Smith KD, Kuehl WM and Bergsagel PL. (1998b). *Blood*, **92,** 3025–3034.
Colvin JS, Bohne BA, Harding GW, McEwen DG and Ornitz DM. (1996). *Nat. Genet.*, **12,** 390–397.
Cools J, Stover EH, Boulton CL, Gotlib J, Legare RD, Amaral SM, Curley DP, Duclos N, Rowan R, Kutok JL, Lee BH, Williams IR, Coutre SE, Stone RM, DeAngelo DJ, Marynen P, Manley PW, Meyer T, Fabbro D, Neuberg D, Weisberg E, Griffin JD and Gilliland DG. (2003). *Cancer Cell*, **3,** 459–469.
Demetri GD, von Mehren M, Blanke CD, Van den Abbeele AD, Eisenberg B, Roberts PJ, Heinrich MC, Tuveson DA, Singer S, Janicek M, Fletcher JA, Silverman SG, Silberman SL, Capdeville R, Kiese B, Peng B, Dimitrijevic S, Druker BJ, Corless C, Fletcher CD and Joensuu H. (2002). *N. Engl. J. Med.*, **347,** 472–480.
Druker BJ. (2003). *Semin. Hematol.*, **40,** 50–58.
Estey EH, Fisher T, Giles F, Feldman EJ, Ehninger G, Schiller GJ, Klimek VM, Nimer SD, De Angelo DJ, Gilliland DG, Fox E, Wang YF, Rosamilia M, Resta D, Chen P and Stone RM. (2003). *Blood*, **102,** 614a.
Fabbro D, Ruetz S, Bodis S, Pruschy M, Csermak K, Man A, Campochiaro P, Wood J, O'Reilly T and Meyer T. (2000). *Anticancer Drug Des.*, **15,** 17–28.
Golla A, Lichmer P, von Gernet S, Winterpacht A, Fairley J, Murken J and Schuffenhauer S. (1997). *J. Med. Genet.*, **34,** 683–684.
Golub TR, McLean T, Stegmaier K, Carroll M, Tomasson M and Gilliland DG. (1996). *Biochim. Biophys. Acta*, **1288,** M7–M10.
Grand EK, Chase AJ, Heath C, Rahemtulla A and Cross NC. (2004). *Leukemia*, **18,** 962–966.
Heffner Jr LT and Lonial S. (2003). *Drugs*, **63,** 1621–1636.
Keats JJ, Maxwell CA, Taylor BJ, Hendzel MJ, Chesi M, Bergsagel PL, Larratt LM, Mant MJ, Reiman T, Belch AR and Pilarski LM. (2005). *Blood*, **105,** 4060–4069.
Kelly LM, Yu JC, Boulton CL, Apatira M, Li J, Sullivan CM, Williams I, Amaral SM, Curley DP, Duclos N, Neuberg D, Scarborough RM, Pandey A, Hollenbach S, Abe K, Lokker NA, Gilliland DG and Giese NA. (2002). *Cancer Cell*, **1,** 421–432.
Lemoli RM, Martinelli G, Zamagni E, Motta MR, Rizzi S, Terragna C, Rondelli R, Ronconi S, Curti A, Bonifazi F, Tura S and Cavo M. (2000). *Blood*, **95,** 2234–2239.
Li Z, Zhu YX, Plowright EE, Bergsagel PL, Chesi M, Patterson B, Hawley TS, Hawley RG and Stewart AK. (2001). *Blood*, **97,** 2413–2419.

Liu Q, Schwaller J, Kutok J, Cain D, Aster JC, Williams IR and Gilliland DG. (2000). *EMBO J.*, **19,** 1827–1838.
Martinelli G, Terragna C, Zamagni E, Ronconi S, Tosi P, Lemoli RM, Bandini G, Motta MR, Testoni N, Amabile M, Ottaviani E, Vianelli N, de Vivo A, Gozzetti A, Tura S and Cavo M. (2000). *J. Clin. Oncol.*, **18,** 2273–2281.
Naski MC, Wang Q, Xu J and Ornitz DM. (1996). *Nat. Genet.*, **13,** 233–237.
Paramore A and Frantz S. (2003). *Nat. Rev. Drug Discov.*, **2,** 611–612.
Paterson JL, Li Z, Wen XY, Masih-Khan E, Chang H, Pollett JB, Trudel S and Stewart AK. (2004). *Br. J. Haematol.*, **124,** 595–603.
Persons DA, Allay JA, Allay ER, Smeyne RJ, Ashmun RA, Sorrentino BP and Nienhuis AW. (1997). *Blood*, **90,** 1777–1786.
Plowright EE, Li Z, Bergsagel PL, Chesi M, Barber DL, Branch DR, Hawley RG and Stewart AK. (2000). *Blood*, **95,** 992–998.
Richardson P, Hideshima T and Anderson K. (2002). *Biomed. Pharmacother.*, **56,** 115–128.
Ronchetti D, Greco A, Compasso S, Colombo G, Dell'Era P, Otsuki T, Lombardi L and Neri A. (2001). *Oncogene*, **20,** 3553–3562.
San Miguel JF, Blade Creixenti J and Garcia-Sanz R. (1999). *Haematologica*, **84,** 36–58.
Santra M, Zhan F, Tian E, Barlogie B and Shaughnessy Jr J. (2003). *Blood*, **101,** 2374–2376.
Schwaller J, Frantsve J, Aster J, Williams IR, Tomasson MH, Ross TS, Peeters P, Van Rompaey L, Van Etten RA, Ilaria Jr R, Marynen P and Gilliland DG. (1998). *EMBO J.*, **17,** 5321–5333.
Sternberg DW, Tomasson MH, Carroll M, Curley DP, Barker G, Caprio M, Wilbanks A, Kazlauskas A and Gilliland DG. (2001). *Blood*, **98,** 3390–3397.
Stone RM, Klimek V, DeAngelo DJ, Nimer S, Estey E, Galinsky I, Neuberg D, Yap A, Fox EA, Gilliland DG and Griffin J. (2003). *Blood*, **100,** 86a.
Tavormina PL, Shiang R, Thompson LM, Zhu YZ, Wilkin DJ, Lachman RS, Wilcox WR, Rimoin DL, Cohn DH and Wasmuth JJ. (1995). *Nat. Genet.*, **9,** 321–328.
Trudel S, Ely S, Farooqi Y, Affer M, Robbiani DF, Chesi M and Bergsagel PL. (2004). *Blood*, **103,** 3521–3528.
Webster MK and Donoghue DJ. (1996). *EMBO J.*, **15,** 520–527.
Weisberg E, Boulton C, Kelly LM, Manley P, Fabbro D, Meyer T, Gilliland DG and Griffin JD. (2002). *Cancer Cell*, **1,** 433–443.
Yagasaki F, Wakao D, Yokoyama Y, Uchida Y, Murohashi I, Kayano H, Taniwaki M, Matsuda A and Bessho M. (2001). *Cancer Res.*, **61,** 8371–8374.